Week 4 Project Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Course Name: ECE501 Digital Image Processing (DIP)

Group Number: Group 15

Group Members: Rinal Shah, Varun Hotani, Aadi Mehta, Pratik Agrawal

Week Number: Week 4

Date: October 11, 2026

Deliverable: Mid-Semester Report, Seven-Fabric Re-Benchmark & Presentation

Objectives of the Week

- Re-run the full pipeline on a broader test subset covering all seven AFID fabric types, and report patch-level and pixel-level metrics under one fixed protocol.
- Audit the evaluation code and data handling (image–mask pairing, defect-name mapping, patch-labelling rule) before reporting any numbers.
- Prepare the 2-page IEEE mid-semester report and the presentation scheduled for 12 October 2026.
- Review the Week 3 follow-ups (Gabor filters, nested 64 × 64 refinement, baseline comparisons) against the mid-semester deadline and decide what is carried forward.

Work Completed This Week

A. Seven-Fabric Re-Benchmark

The pipeline from the reference implementation (commit 9c9fa70) was re-run on 13 AFID images spanning all seven fabric IDs (00–06) and six defect codes (002, 006, 010, 016, 019, 027). Patch-level and pixel-level metrics were computed per image and then averaged. This extends the Week 3 benchmark, which covered only three weave types, to every fabric in the dataset. The older repository's synthetic results were excluded so that only real AFID data is reported.

B. Pipeline and Evaluation Audit

Before reporting results, the data-handling and evaluation code was inspected. The committed settings were recorded: 3 × 3 median filter, CLAHE with clip limit 2.0 on an 8 × 8 tile grid, 256 × 256 patches at stride 128, covariance regularisation $\Sigma + \varepsilon \cdot \mathrm{tr}(\Sigma) \cdot I$ with $\varepsilon = 10^{-4}$ and a pseudoinverse, per-fabric threshold $\tau = \mu + k\sigma$ with $k = 2.5$, 5 × 5 opening and closing, and removal of components below 50 pixels. A patch is labelled defective when it holds more than 25 defect pixels. Pixel F1 equals Dice under matching mask conventions.

The audit found three issues. The loader's defect-name mapping was incorrect, so defect names in the results were corrected to the official AFID defect codes. The loader can replace a missing ground-truth mask with an all-zero mask, which can silently turn a missed pairing into a zero-overlap result. The

training counts, splits and ROI handling of the latest run are not documented, so the committed settings cannot yet be confirmed as the exact configuration of that run.

C. Mid-Semester Report and Presentation

The 2-page IEEE mid-semester report and the 8-slide presentation (12 October 2026) were written from the re-benchmark. Both state the limits of the evidence plainly: the small test subset, the unverified weave-name mapping for fabric IDs, and the pending mask checks.

D. Scope Decisions for the Remaining Weeks

Gabor filter banks were moved to optional future work. The nested 64 × 64 refinement, the controlled fixed-versus-adaptive parameter comparison and the baseline comparisons planned in Week 3 were not completed this week and are carried forward.

Literature Review

Topic/Paper	Key Takeaway	Relevance to the Project
Textural Features for Image Classification [1]	Defines gray-level co-occurrence statistics (contrast, dissimilarity, homogeneity, energy, correlation) as compact texture descriptors.	Basis of the GLCM module; angular mean and standard deviation give 30 values per patch.
Multiresolution Gray-Scale and Rotation Invariant Texture Classification with LBP [2]	Defines uniform, rotation-invariant LBP histograms at several (P, R) settings.	Source of the two LBP scales used, (8, 1) and (16, 2); also the basis for the planned fixed-versus-adaptive scale comparison.
A Public Fabric Database for Defect Detection Methods and Results [3]	Describes the 245-image AFID benchmark, its 12 defect classes, and the pixel-level ground-truth masks.	Defines the dataset, defect codes and mask format used to re-label results and check image–mask pairing.
On the Generalized Distance in Statistics [4]	Mahalanobis distance measures how far a sample lies from a distribution, accounting for feature scale and correlation.	Anomaly score against each fabric's normal-texture model.
Unsupervised Texture Segmentation Using Gabor Filters [5]	Multi-orientation, multi-frequency Gabor filter banks separate texture by orientation and scale.	Candidate extra descriptor, now optional future work.

Quantitative Results and Analysis

The re-benchmark covers 13 AFID images across seven fabric IDs and six defect codes. Patch metrics describe whether defective 256 × 256 regions are found; pixel metrics describe how well the flagged area matches the ground-truth mask. All values are means of per-image metrics.

Table 1. Mean per-image performance on the Week 4 subset (%).

Level	Accuracy	Precision	Recall	F1-score	IoU
Patch	79.83	21.88	41.54	27.13	20.56
Pixel	80.66	0.36	43.57	0.72	0.36

Table 2. Fabric-wise results (%). Fabric IDs are used because the weave-name mapping is unverified; n is the image count.

Fabric ID	n	Patch accuracy	Patch F1	Pixel IoU
00	2	95.55	80.00	1.16
01	2	74.19	18.18	0.48
02	2	70.97	18.18	0.28
03	2	78.26	20.00	0.34
04	2	74.19	0.00	0.00
05	1	96.55	80.00	0.20
06	2	77.42	0.00	0.00

Table 3. Defect-wise means derived from the sample rows, using official AFID defect codes (%).

Code	Defect	n	Patch F1	Pixel IoU
002	Broken end	4	49.09	0.82
006	Broken yarn	2	18.18	0.28
010	Broken pick	1	40.00	0.68
016	Weft curling	1	0.00	0.00
019	Fuzzy ball	4	0.00	0.00
027	Knot	1	80.00	0.20

Discussion of Findings

1. **Patch detection is modest on the full fabric range.** Mean patch accuracy is 79.83% with F1 of 27.13%. Precision (21.88%) is lower than recall (41.54%), so many flagged patches are false alarms and over half of defective patches are missed.
2. **Pixel accuracy hides weak localization.** Pixel accuracy of 80.66% is driven by normal background. Pixel precision is 0.36%, F1 (equal to Dice) is 0.72% and IoU is 0.36%. Assigning a score to a whole 256 × 256 window (65,536 pixels) includes much normal area around a small defect, which is a plausible contributor.
3. **Fabric and defect results are uneven.** Fabric IDs 00 and 05 reach the best patch F1 (80%), but their pixel IoU stays near 1% or below. Fabrics 04 and 06 show zero overlap, which needs mask checks before it is read as a detection failure. Weft curling and fuzzy ball score 0% F1.

4. **These Week 4 figures supersede the Week 3 table.** Week 3 reported 95.39% patch accuracy and 100% pixel recall on ten strips covering three weave types. Week 4 covers all seven fabrics and gives 79.83% and 43.57%. The two sets differ in coverage and in how complete the configuration is, so they are not directly comparable, and the Week 4 subset is the more representative of the two. The 100% pixel recall in Week 3 was partly a side effect of patch-sized flagging.
5. **The improvement over the earlier real-data run is not yet attributable to adaptation.** Patch accuracy rose from roughly 18–20% in earlier real-data runs to 79.83%, but coverage and configuration also changed. The controlled fixed-versus-adaptive comparison is still pending.
6. **Evidence is thin.** Each fabric has only one or two images and each defect code one to four, so fabric and defect effects are confounded. All figures are preliminary.

Challenges Faced

- **Results that could not be compared across weeks.** The Week 3 benchmark used ten strips on three weave types, while Week 4 uses 13 images on all seven fabrics. The headline numbers moved sharply, so the earlier table had to be explicitly superseded rather than extended.
- **Data-handling defects in the loader.** Incorrect defect-name mappings and the substitution of an all-zero mask for a missing ground truth can distort results. Fabrics 04 and 06 show zero overlap, and until pairing is verified it is unclear whether that is a real miss.
- **Undocumented run configuration.** The training counts, splits and ROI handling of the latest run are not recorded, so committed settings cannot yet be confirmed as the run's exact configuration.
- **Coarse localization.** A thin yarn flaw of roughly 1,500 pixels flags a 65,536-pixel patch, which caps pixel IoU at about 2% and keeps pixel precision near 0.4%.
- **Small and imbalanced evidence.** One to two images per fabric and one to four per defect code make fabric-wise and defect-wise conclusions unreliable.
- **Unverified fabric naming.** The mapping from fabric IDs to weave names is unconfirmed, so results are reported by ID only.
- **Time pressure.** Preparing the mid-semester report and presentation left no time for the 64 × 64 refinement or the baseline comparisons.

Plan for Next Week (Week 5)

- **Validate ground truth.** Verify image–mask pairing and the official defect labels, inspect the zero-overlap cases on fabrics 04 and 06, and remove the all-zero-mask fallback.
- **Hierarchical refinement.** Refine flagged 256 × 256 regions with nested 64 × 64 sub-patches to raise pixel IoU above the current 0.36%.
- **Controlled comparisons.** Compare fixed and adaptive patch sizes and GLCM/LBP settings under identical image splits, with GLCM-only, LBP-only and combined descriptors.
- **Larger test set.** Evaluate more images, including defect-free ones, to measure false alarms and image-level decisions.
- **Fuller reporting.** Report defect-type-wise and image-level performance, save per-image masks and overlays, and evaluate detection reliability scores.
- **Reproducibility.** Document the training counts, splits, ROI handling and full configuration of each reported run, and commit the code and results.
- **If time allows.** Gabor filter banks and One-Class SVM / Isolation Forest baselines.

Conclusion

Week 4 moved the project from an initial three-weave benchmark to a seven-fabric re-benchmark and produced the mid-semester report and presentation. On 13 AFID images the pipeline reaches 79.83% patch accuracy and 27.13% patch F1, with pixel IoU of only 0.36%, so patch-level detection is demonstrated but precise localization is unresolved. The audit found loader defects and an undocumented run configuration, and these figures replace the Week 3 table. Week 5 therefore focuses on validating the masks, nested 64 × 64 refinement and controlled comparisons before the end-semester stage.

Citations

[1] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," IEEE Trans. Syst., Man, Cybern., vol. SMC-3, no. 6, pp. 610–621, Nov. 1973. doi: 10.1109/TSMC.1973.4309314.

[2] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," IEEE Trans. Pattern Anal. Mach. Intell., vol. 24, no. 7, pp. 971–987, Jul. 2002. doi: 10.1109/TPAMI.2002.1017623.

[3] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A public fabric database for defect detection methods and results," AUTEX Res. J., vol. 19, no. 4, pp. 363–374, Dec. 2019. doi: 10.2478/aut-2019-0035.

[4] P. C. Mahalanobis, "On the generalized distance in statistics," Proc. Nat. Inst. Sci. India, vol. 2, no. 1, pp. 49–55, 1936.

[5] A. K. Jain and F. Farrokhnia, "Unsupervised texture segmentation using Gabor filters," Pattern Recognit., vol. 24, no. 12, pp. 1167–1186, 1991. doi: 10.1016/0031-3203(91)90143-S.